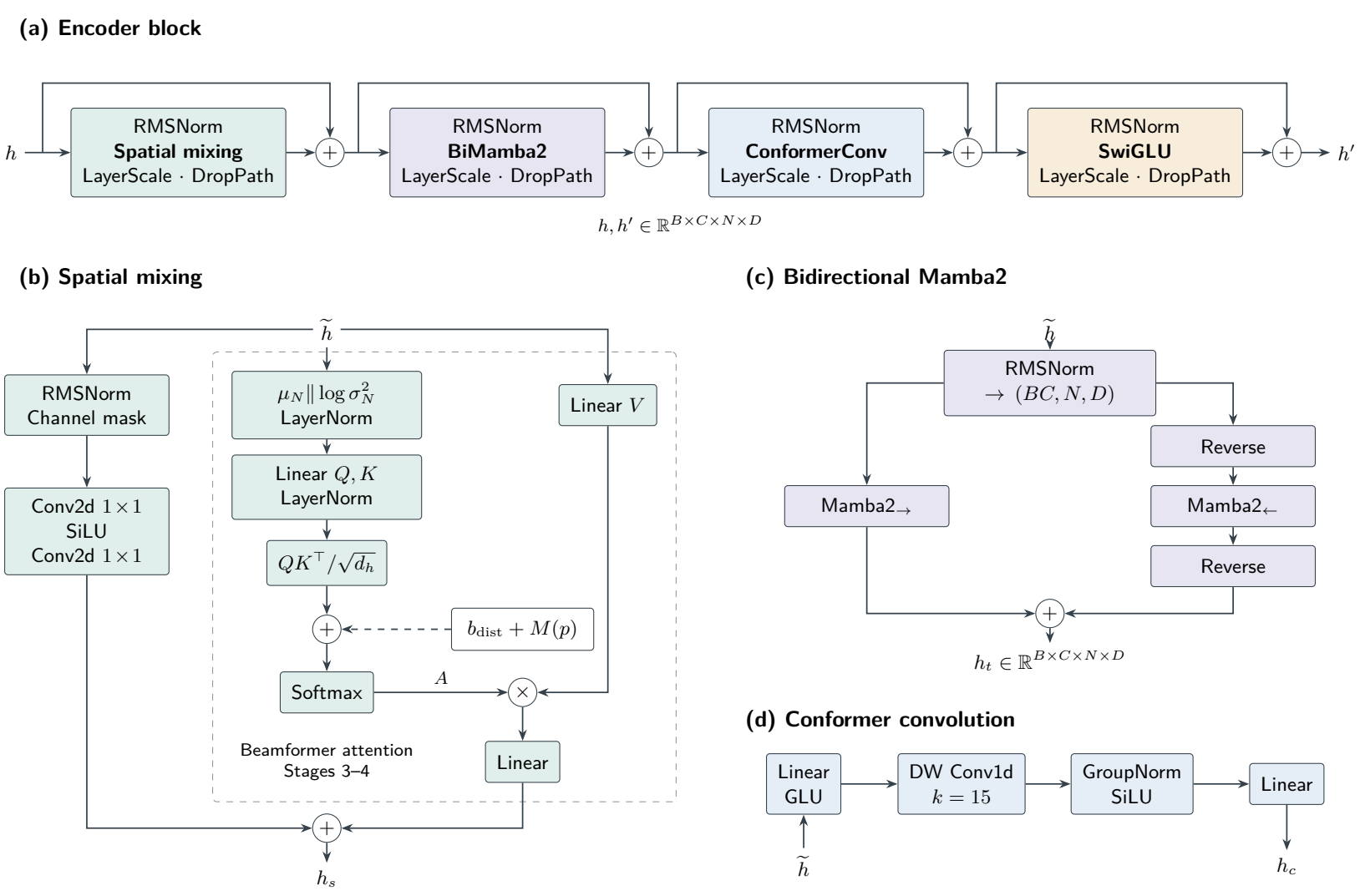

(a) Encoder block
h
RMSNorm
Spatial mixing
LayerScale · DropPath
RMSNorm
BiMamba2
LayerScale · DropPath
RMSNorm
ConformerConv
LayerScale · DropPath
RMSNorm
SwiGLU
LayerScale · DropPath
h′
h, h′ ∈ ℝ^{B×C×N×D}
(b) Spatial mixing
h̃
RMSNorm
Channel mask
Conv2d 1×1
SiLU
Conv2d 1×1
μ_N ‖ log σ²_N
LayerNorm
Linear Q, K
LayerNorm
QK^⊤/√d_h
b_dist + M(p)
Softmax
A
Linear V
Beamformer attention
Stages 3–4
Linear
h_s
(c) Bidirectional Mamba2
h̃
RMSNorm
→ (BC, N, D)
Reverse
Mamba2→
Mamba2←
Reverse
h_t ∈ ℝ^{B×C×N×D}
(d) Conformer convolution
Linear
GLU
DW Conv1d
k = 15
GroupNorm
SiLU
Linear
h̃
h_c